



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-15274
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2786.93
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10494993	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 15274
CALIBRATION DATE: 12-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

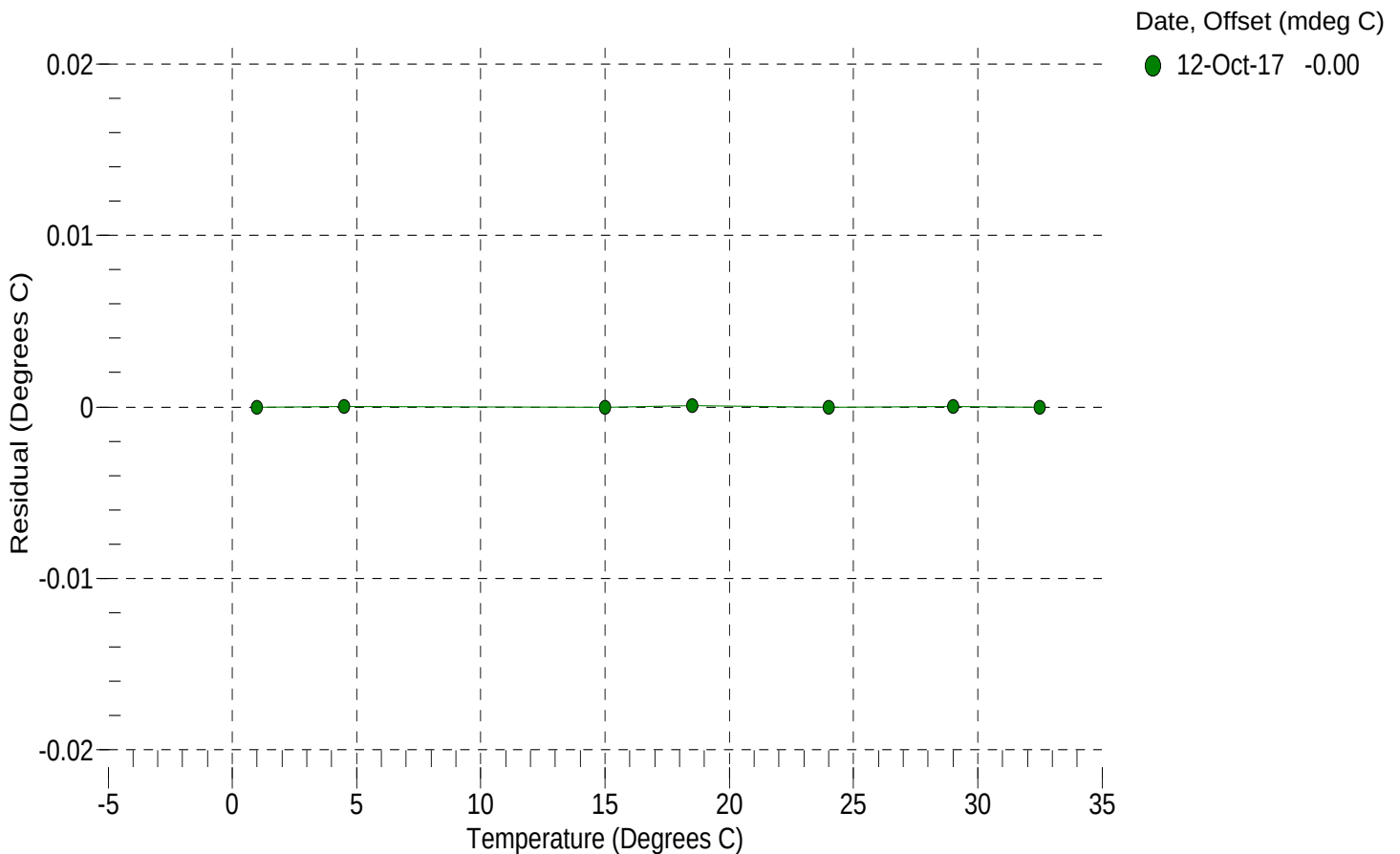
a0 = -2.270447e-004
a1 = 3.187868e-004
a2 = -4.777132e-006
a3 = 2.104011e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	568082.6	1.0000	-0.0000
4.4999	487968.3	4.4999	0.0000
15.0000	315269.2	15.0000	-0.0000
18.5000	274219.6	18.5001	0.0001
24.0000	221535.4	24.0000	-0.0000
29.0000	183571.6	29.0000	0.0000
32.5000	161464.9	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -1.011634e+000
h = 1.304654e-001
i = -1.523659e-004
j = 2.672078e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 8.9699e-008

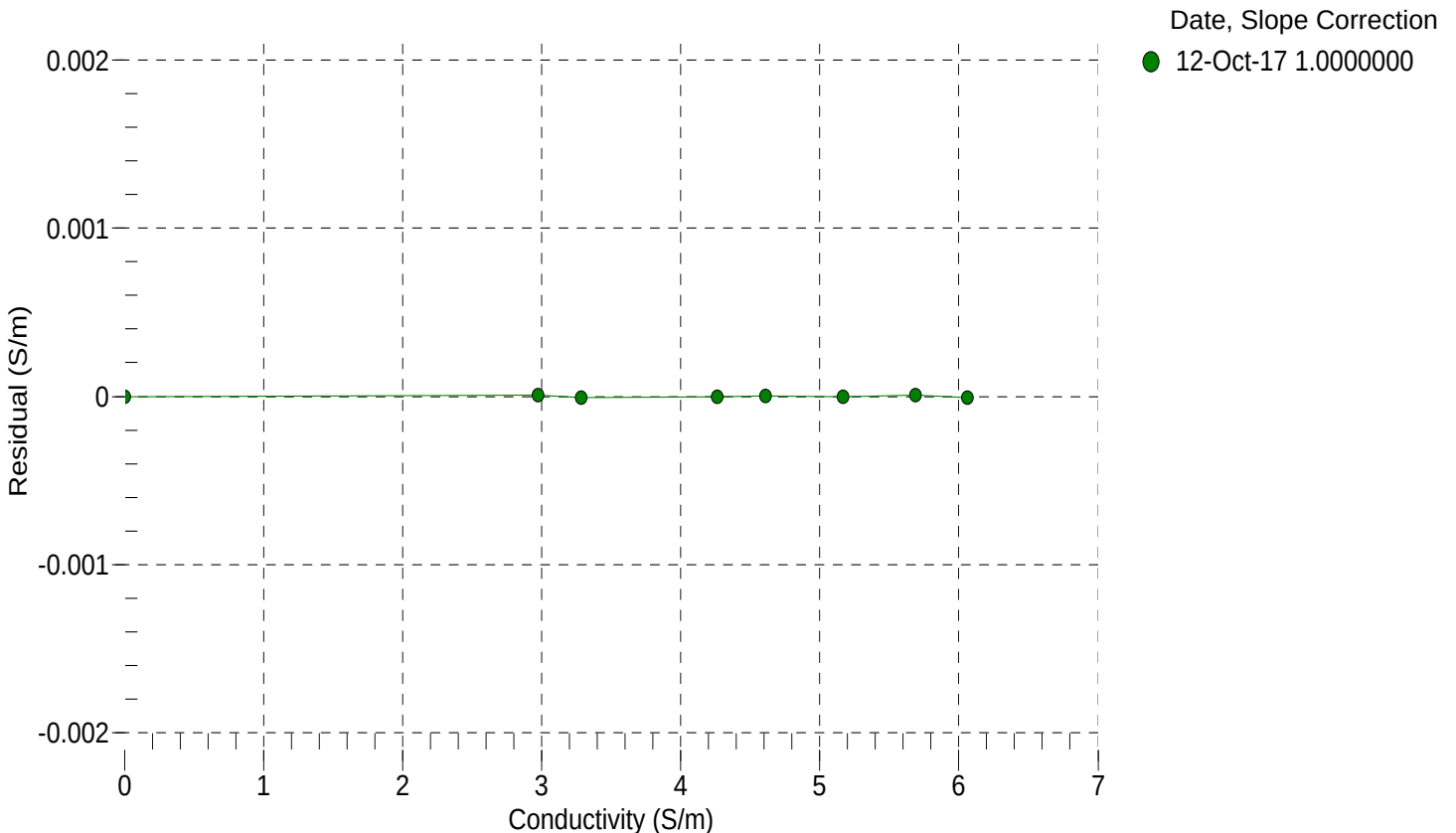
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2786.93	0.00000	0.00000
1.0000	34.8183	2.97612	5529.17	2.97613	0.00001
4.4999	34.7982	3.28318	5737.43	3.28317	-0.00001
15.0000	34.7542	4.26480	6356.90	4.26480	-0.00000
18.5000	34.7448	4.60991	6560.55	4.60991	0.00000
24.0000	34.7340	5.16774	6876.69	5.16773	-0.00000
29.0000	34.7277	5.68944	7159.42	5.68945	0.00001
32.5000	34.7240	6.06171	7354.34	6.06171	-0.00001

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 15274
CALIBRATION DATE: 24-Oct-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10494993

COEFFICIENTS:

PA0 =	1.456928e-002	PTCA0 =	5.242497e+005
PA1 =	9.008052e-003	PTCA1 =	2.452289e+000
PA2 =	-1.641827e-010	PTCA2 =	-1.127323e-001
PTEMPA0 =	-6.398386e+001	PTCB0 =	2.527950e+001
PTEMPA1 =	5.269211e-002	PTCB1 =	-3.000000e-004
PTEMPA2 =	-5.672343e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.78	525893.2	1652.3	14.82	0.00	32.50	1869	525902.80
592.02	590029.6	1654.6	592.00	-0.00	29.00	1800	525924.75
1169.04	654296.9	1656.1	1169.01	-0.00	24.00	1701	525943.61
1746.12	718727.5	1657.0	1746.12	-0.00	18.50	1593	525952.32
2323.11	783297.3	1658.0	2323.11	0.00	15.00	1524	525952.95
2900.13	848020.7	1659.3	2900.10	-0.00	4.50	1319	525954.24
2323.12	783300.4	1658.6	2323.14	0.00	1.00	1250	525949.11
1746.06	718725.4	1658.5	1746.11	0.00	TEMPERATURE (°C) SPAN (mV)		
1169.12	654306.0	1658.4	1169.10	-0.00			
592.10	590040.3	1658.7	592.11	0.00			
14.77	525884.6	1658.9	14.75	-0.00	-5.00		25.28
					35.00		25.27

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

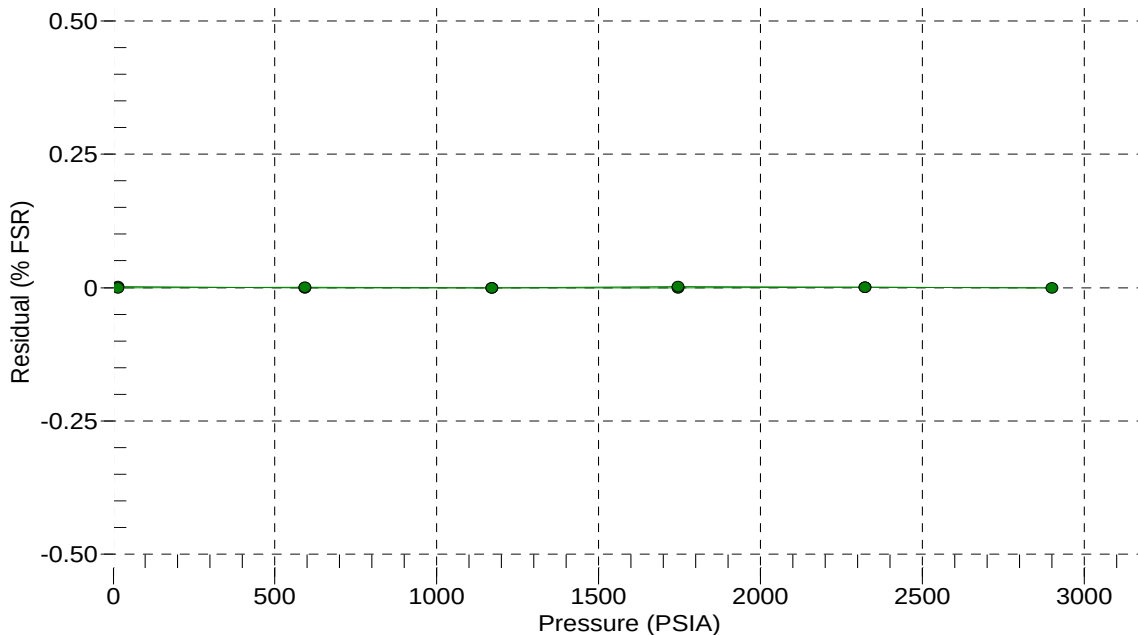
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 24-Oct-17 -0.00





Sea-Bird Electronics, Inc.

13431 NE 20th St. Bellevue, Washington 98005 USA
www.seabird.com

Phone: (425) 643-9866

Fax: (425) 643-9954

Email: seabird@seabird.com

Pressure Test Certificate

Test Date: **2016-11-22**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **15274**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **pch**

High pressure is generally equal to the maximum depth rating of the instrument

